

A. Lyapunov gauge decomposition

$$J(x) = (-D/2 + A)\Sigma^{-1}$$

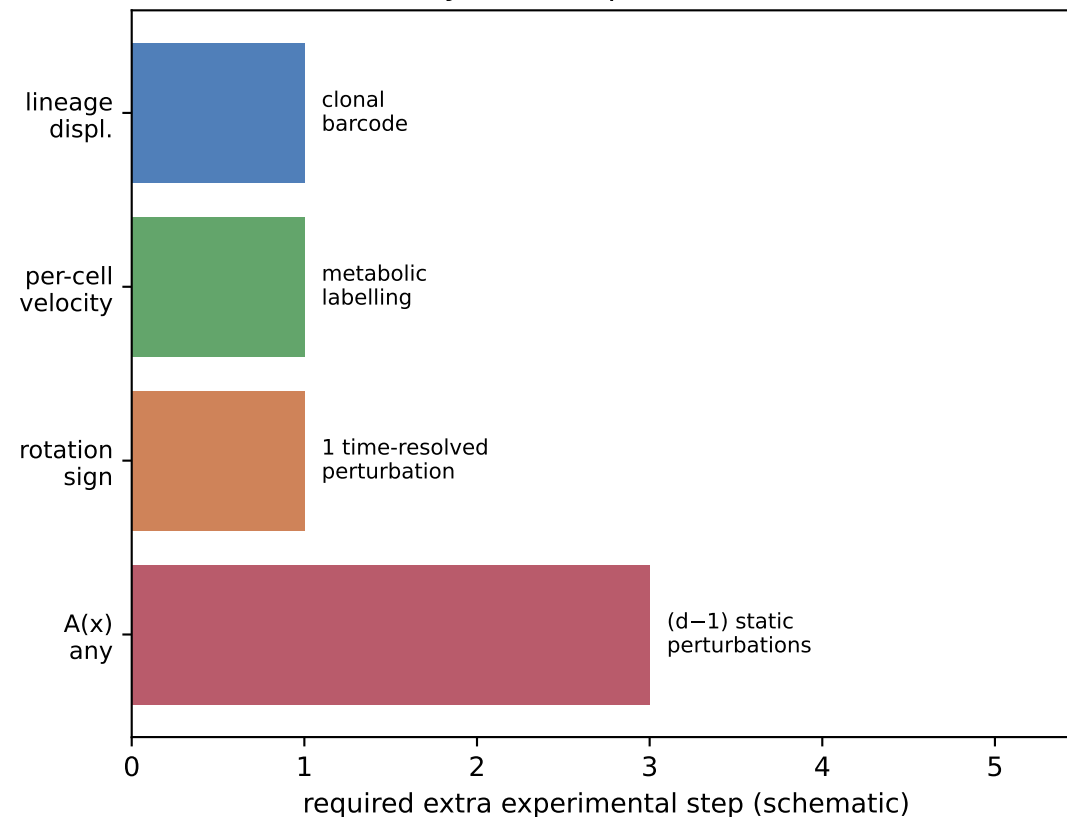
symmetric part: covariance-derived, identifiable

$$\text{sym}(\Sigma^{-1/2}J\Sigma^{1/2}) = -\frac{1}{2}\Sigma^{-1/2}D\Sigma^{-1/2}$$

antisymmetric part A : not identified

flow direction, oscillation phase, saddle timing

B. Price of the antisymmetric part



C. Estimator on marrow Ery (Section 2.1)

	median cos	Frob corr
J vs local covariance	0.16 [EL01]	—
J vs local precision	0.36 [EL02]	—
sym_W vs $-\frac{1}{2}\Sigma^{-1/2}D\Sigma^{-1/2}$	0.454 [EL03]	+0.468
sym_W vs local precision	0.224 [EL03]	−0.368